BIG IDEAS:

- A **set** is a collection of objects, such as numbers, in which order does not matter
- A **subset** is made up of objects from another set, called a **parent set**
- The set of **real numbers** has many subsets (whole numbers, integers, rational numbers, etc.) that are used when solving problems in various contexts

EXAMPLE 1

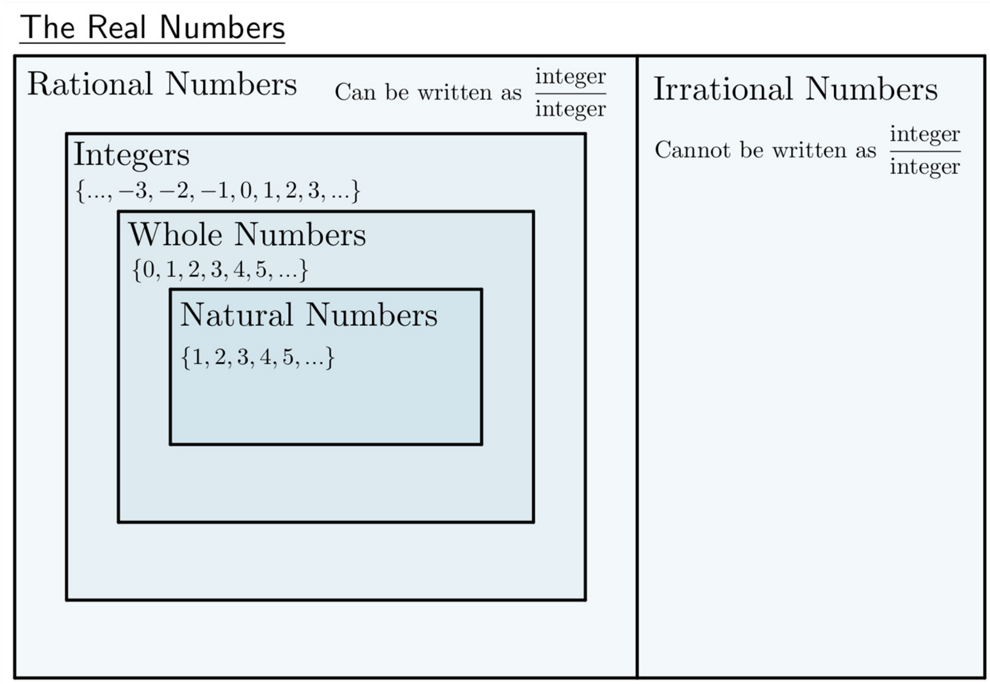
Some subsets of the set $\{1, 3, 5, 7, 9\}$ are $\{1, 5, 9\}$, $\{3,7\}$ and $\{\ \}$.

EXAMPLE 2

The set of odd integers from -5 through 5 is $\{-5,-3,-1,1,3,5\}$.

EXAMPLE 3

The set of whole numbers is a subset of the set of real numbers.



MORE EXAMPLES

Things to remember…

THINKING BEYOND…

1. Sets can have other sets as elements! The **power set** of a set A is the set containing all the subsets of set A. Determine the power set of $\{1,3,5,7\}$.
2. In a school, there are currently several students studying English, French, or both of these languages. Given that there are 784 students studying English, 367 students studying French, and 210 students studying both languages, how many students are studying languages in total?
3. How many different subsets of $\{1,2,3,4,\ldots,100\}$ can be made containing exactly 10 elements?